ELECIFY

BY GROUP 10 STRATEGIST

PAINPOINT

SOMETIMES PEOPLE UNKNOWINGLY WASTE ELECTRICITY, ONLY REALIZING IT WHEN THE NEXT MONTHLY BILL ARRIVES—AT WHICH POINT IT IS STILL IMPOSSIBLE TO BREAK DOWN THE POWER CONSUMPTION BY INDIVIDUAL HOUSEHOLD APPLIANCES.

ROOT CAUSE

THE LACK OF REAL-TIME POWER MONITORING AND THE UNAVAILABILITY OF ITEMIZED, APPLIANCE-LEVEL ENERGY BREAKDOWN



TARGET AUDIENCE

DORMITORY RENTERS / STUDENTS

WHO NEED CLEAR VISIBILITY OF DAILY/MONTHLY ELECTRICITY COSTS AND APPLIANCE CONSUMPTION BREAKDOWNS TO BUDGET EFFECTIVELY.

HOME / PROPERTY LEADERS

WHO WANT TO MANAGE ROOM ELECTRICITY USAGE HABITS AND MAINTAIN ACCURATE MONTHLY EXPENSE ESTIMATES.

DORMITORY LANDLORDS / UTILITY PROVIDERS

WHO BENEFIT FROM TRANSPARENT BILLING UNDERSTANDING BY RENTERS TO AVOID RATE DISPUTE ISSUES.

Core Feature

- Profile to make preset for user
- Computing system - Calculating the result from inputs
  - Calculation formula used 

$$\text{kWh} = \frac{\text{Watts} \times \text{Operating Hours}}{1000}$$

- User guide – displays instuctions and help documentation to assist users in operating the system correctly
- Result decare - display total result for all electric appliances in the list
- Languages and theme - allows users to toggle between Thai/English and switch UI themes for optimal usability

NON FUNCTIONAL REQUIREMENT

- NFR-1 (PERFORMANCE): DISPLAY UPDATED TOTAL RESULTS WITHIN 1.5 SECONDS AFTER PRESSING "CALCULATE".
- NFR-2 (PERFORMANCE): COMPLETE PROFILE SETUP AND FIRST CALCULATION WITHIN 45 SECONDS WITHOUT EXTERNAL HELP.
- NFR-3 (USABILITY): PROFILE AND HISTORY DATA PERSIST WITHOUT CORRUPTION ACROSS RESTARTS.
- NFR-4 (RELIABILITY): 100% MATHEMATICAL ACCURACY USING FORMULA: COST = (WATTAGE * HOURS) / 1000 * UNIT RATE.
- NFR-5 (SECURITY/PRIVACY): USER PROFILE DATA AND CALCULATION LOGS STRICTLY ISOLATED PER USER SESSION.